

BIOMECHANICS (EPHE 341)

Work, energy and power

McGinnis Chpt 4

By the end of this lecture you should be able to...

- **Define** mechanical work, energy and power, and state their SI units
- **Distinguish** kinetic, gravitational and elastic energy in movement
- **Connect** the force–velocity–power curve to performance
- **Apply** the work–energy theorem to jumping, lifting and running
- **Use** conservation of mechanical energy to solve problems
- **Interpret** the stretch–shortening cycle in energetic terms

A second way of describing the same motion

FORCE AND MOMENTUM — what you have already seen

- Newton's second law: $\Sigma F = m \cdot a$
- Impulse–momentum: $F \cdot \Delta t = \Delta(m \cdot v)$
- Tracks force and how it changes motion **over time**

WORK AND ENERGY — what we add today

- Work: force acting **over a distance**
- Energy: the capacity to do work — kinetic, gravitational, elastic
- Power: the **rate** at which work is done

Impulse is force integrated over **time** and gives you velocity. Work is force integrated over **distance** and gives you energy. Same force record, two different integrals, two different questions answered.

A scalar — force acting over a displacement

$$W = F \cdot d \cdot \cos \theta$$

SI unit

joule (J)

$$1 \text{ J} = 1 \text{ N} \cdot \text{m} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$$

A scalar

No direction of its own — the sign comes entirely from $\cos \theta$, the angle between the force and the displacement

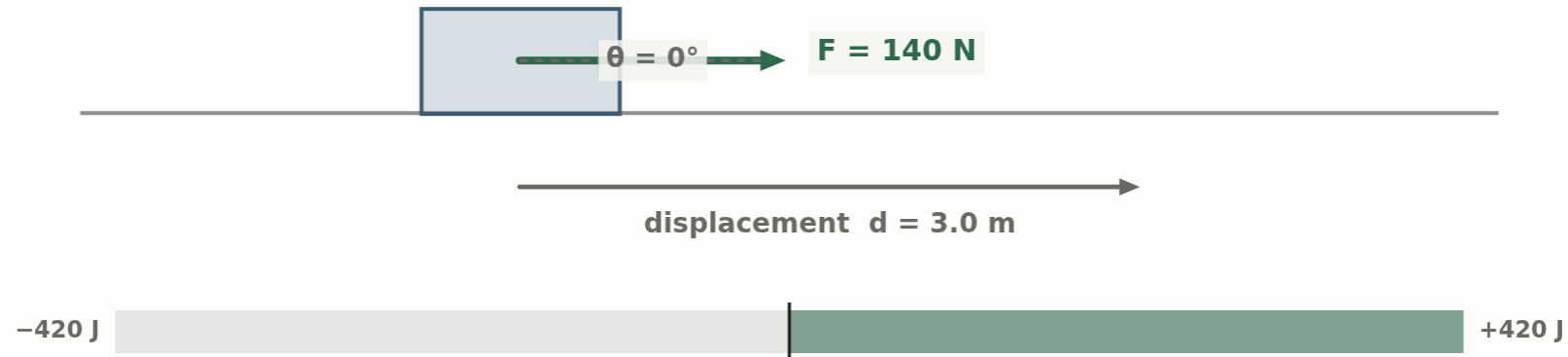
Net work

Sum over every force acting, or equivalently $W_{\text{net}} = F_{\text{net}} \cdot d \cdot \cos \theta$

The sign of work

Turn the force and watch $\cos \theta$ change the answer

POSITIVE WORK



$$W = F \cdot d \cdot \cos \theta = 140 \times 3.0 \times \cos 0^\circ = \mathbf{420 \text{ J}} \cdot \cos \theta = 1.00$$

Angle θ

Distance

Force

Example Problem

An athlete lifts a 100 kg barbell 2.1 m from the floor to overhead at a roughly constant velocity. How much work does the athlete do on the bar?

Given:

$$m = 100 \text{ kg}$$

$$d = 2.1 \text{ m (vertical)}$$

$$a \approx 0 \text{ (constant velocity)}$$

$$\theta = 0^\circ \text{ (force and displacement both up)}$$

Solution

- At constant velocity $\Sigma F = 0$, so the lifting force equals the weight of the bar

$$F = m \cdot g = (100 \text{ kg})(9.81 \text{ m/s}^2)$$

$$F = 981 \text{ N}$$

- The force and the displacement are both upward, so $\theta = 0^\circ$ and $\cos \theta = 1$

$$W = F \cdot d \cdot \cos(0^\circ)$$

$$W = (981 \text{ N})(2.1 \text{ m})(1)$$

$$W \approx 2060 \text{ J}$$

The bar's gravitational potential energy has gone up by the same 2060 J.

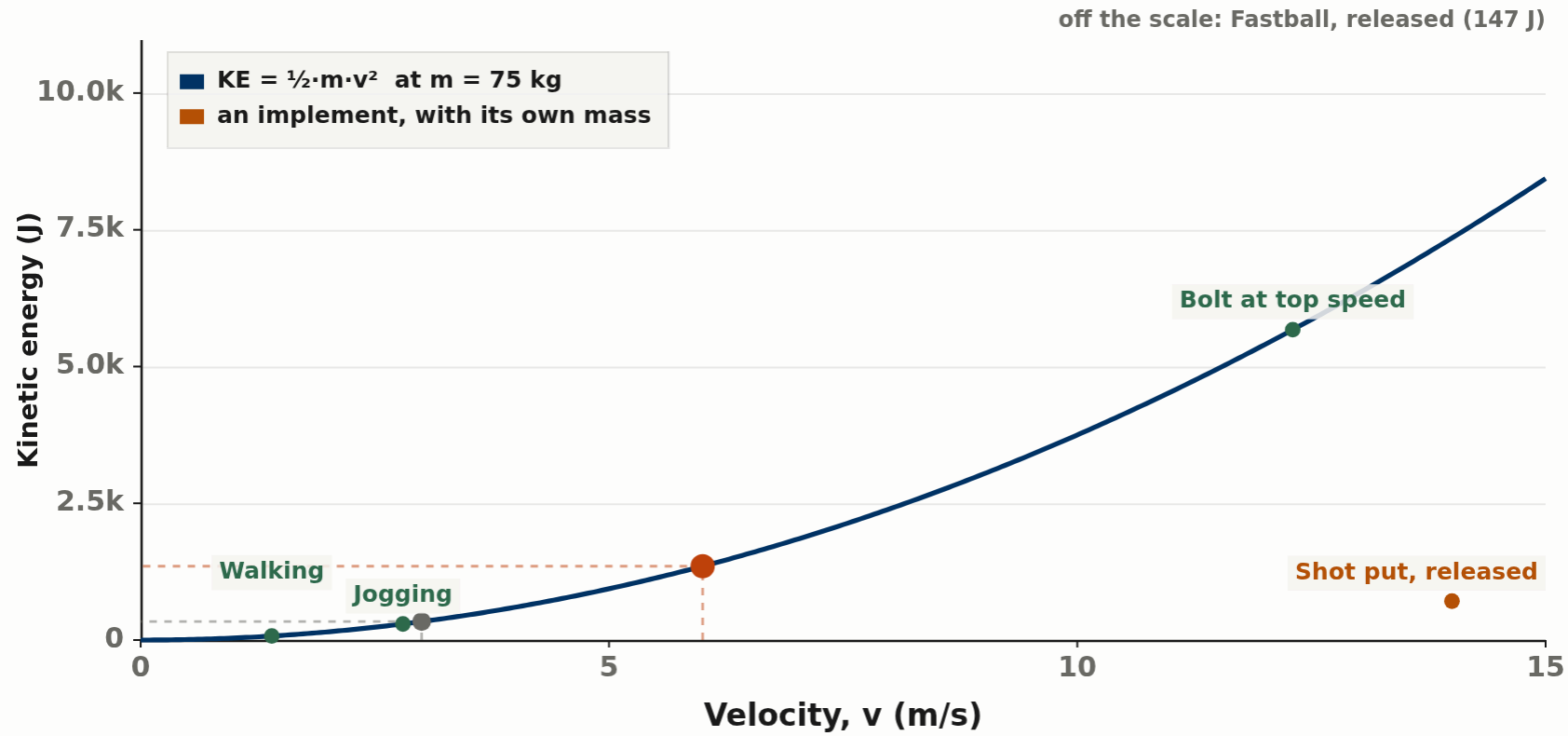
The energy an object has because it is moving

$$KE = \frac{1}{2} \cdot m \cdot v^2$$

- A scalar, and always ≥ 0
- **Doubling the speed quadruples it**
- Units: joules, the same as work
- Halving the mass only halves it

Kinetic energy

Doubling the velocity quadruples the energy



at **3.0 m/s** → 338 J · at **6.0 m/s** → **1350 J** — twice the speed, 4.0× the energy

set the speed:

walking

jogging

sprinting

The v^2 term is what makes these numbers spread

Bolt at top speed

$m \approx 86 \text{ kg}$, $v \approx 12.3 \text{ m/s}$
 $\approx 6,500 \text{ J}$

Shot put at release

$m = 7.26 \text{ kg}$, $v \approx 14 \text{ m/s}$
 $\approx 710 \text{ J}$

70 kg runner, jogging

$v \approx 2.8 \text{ m/s}$
 $\approx 275 \text{ J}$

Fastball at release

$m \approx 145 \text{ g}$, $v \approx 45 \text{ m/s}$
 $\approx 150 \text{ J}$

A 145 g baseball carries more than half the kinetic energy of a jogging adult, because it is moving sixteen times faster. Mass differs by a factor of 480; the energies differ by a factor of under two.

Net work equals the change in kinetic energy

$$W_{\text{net}} = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

WHAT IT SAYS

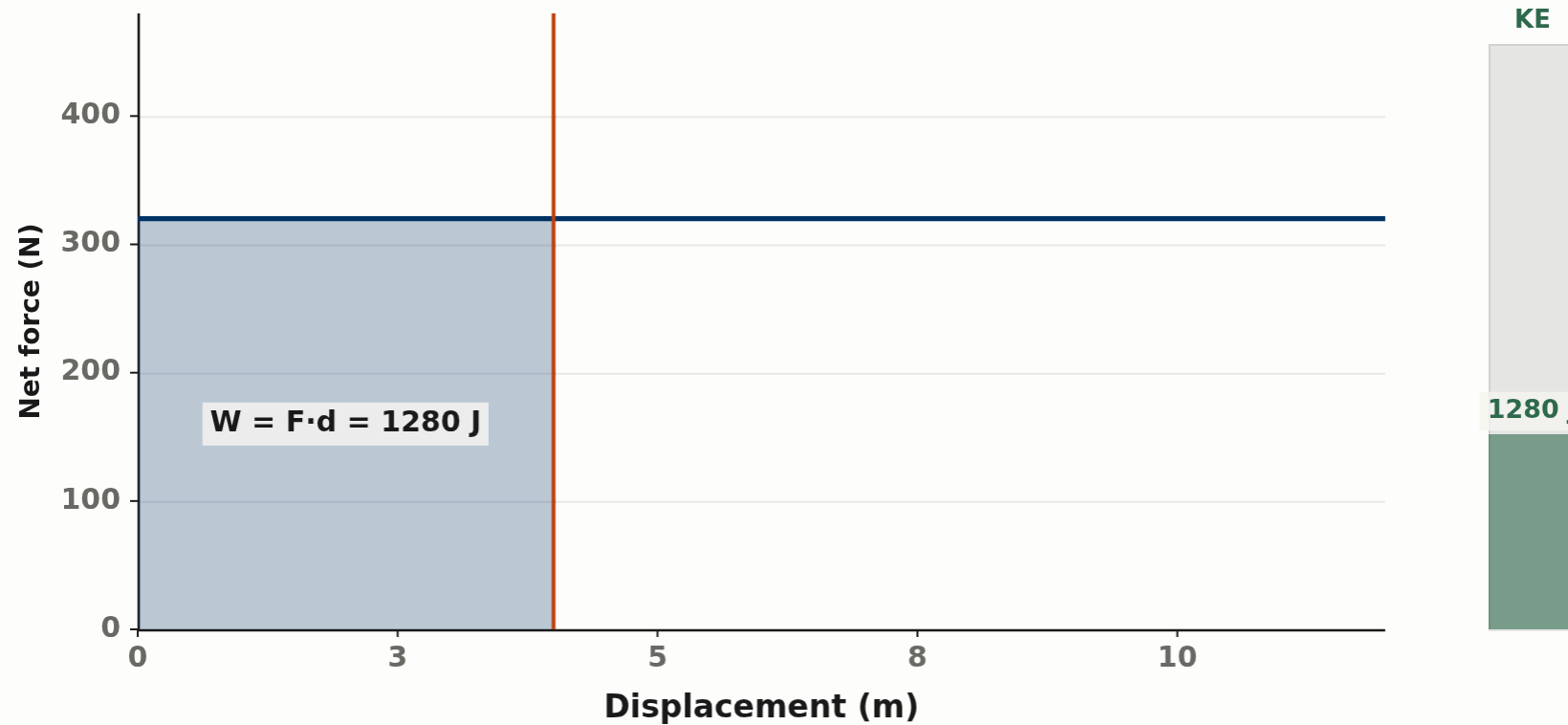
- $W_{\text{net}} > 0 \rightarrow$ the object speeds up
- $W_{\text{net}} < 0 \rightarrow$ it slows down
- $W_{\text{net}} = 0 \rightarrow$ the speed is unchanged

WHERE IT IS USED

- Sprint acceleration: net forward work becomes kinetic energy
- Braking at heel-strike: negative work removes it again
- Throwing: work on the implement becomes its kinetic energy

The work–energy theorem

The area and the energy are the same number

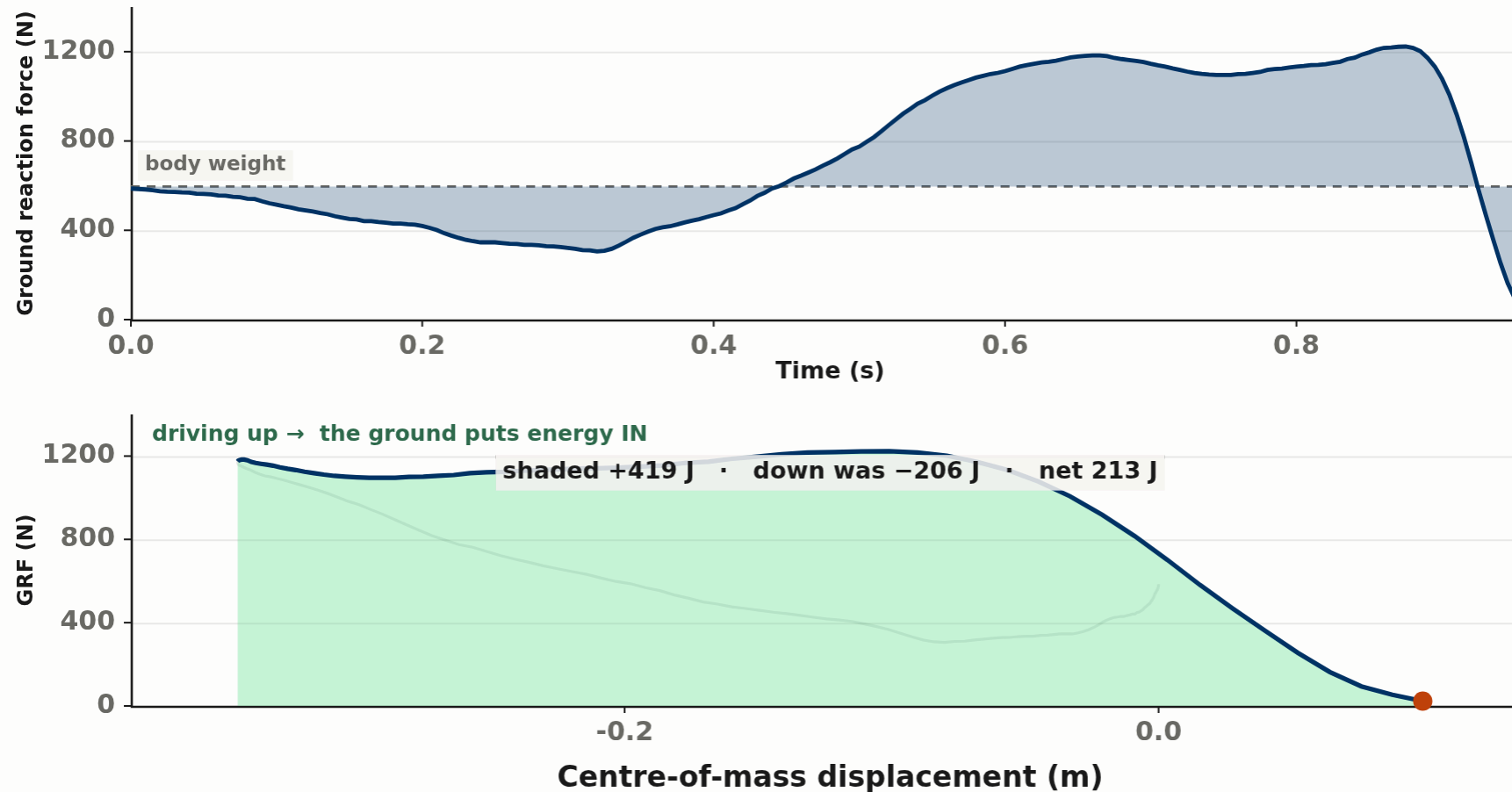


$$W_{\text{net}} = 1280 \text{ J} \cdot \Delta KE = \frac{1}{2}(75)(5.84)^2 = 1280 \text{ J} \cdot v = 5.84 \text{ m/s after 4.0 m}$$

Net force Distance run Mass

Work in a vertical jump

The measured jump from Linear Kinetics II, integrated once more



GRF work so far **213 J** = ΔKE **154 J** + ΔPE **59 J** · $v = 2.25 \text{ m/s}$ · CoM 0.10 m · measured, SUSU-46



Through the
jump

0.96 s

Part 02

Energy forms

Kinetic, gravitational, elastic — and conservation

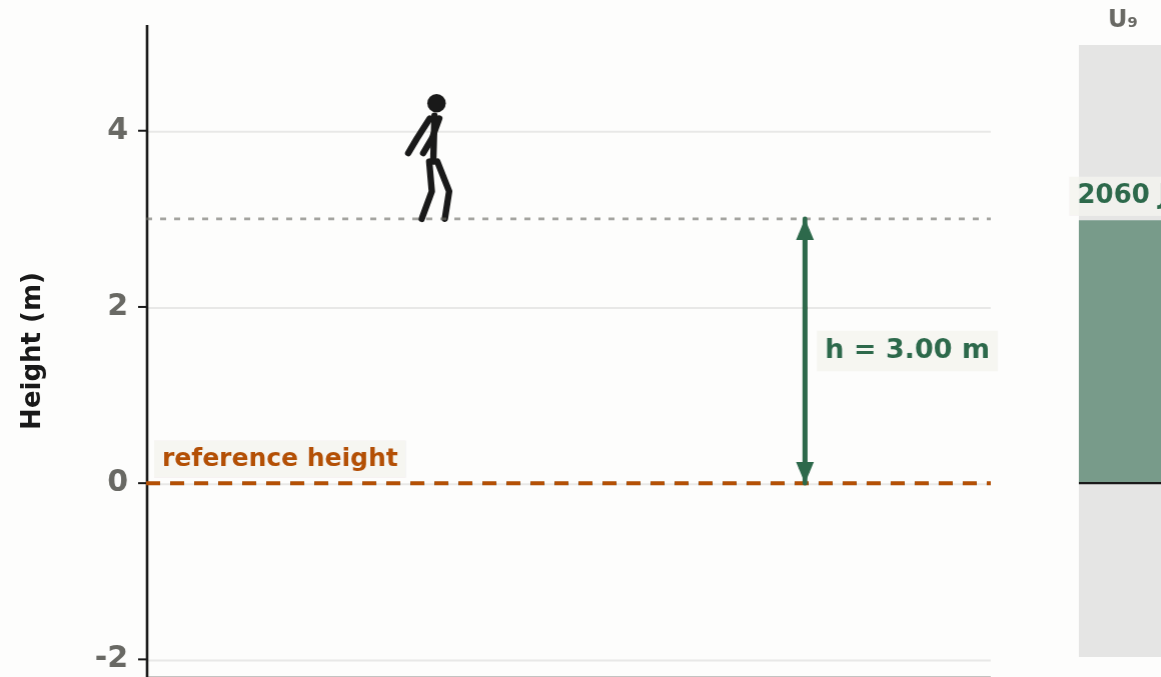
Energy stored in an object's vertical position

$$U_g = m \cdot g \cdot h$$

- **h is measured from a reference height that you choose** — only *changes* in U_g are physically meaningful
- Raising 1 kg by 1 m stores about 9.81 J
- A change in U_g is minus the work done by gravity

Gravitational potential energy

Move the reference and the number changes; the climb does not



$U_g = m \cdot g \cdot h = 70 \times 9.81 \times 3.00 = \mathbf{2060 \text{ J}}$ · move the reference and U changes; the climb does not

Mass **70 kg**

Height **3.0 m**

Reference **0.0 m**

Example Problem

A 70 kg student climbs a staircase 3.0 m high. How much gravitational potential energy do they gain?

And if the climb takes 2.0 s, what average power did they produce?

Given:

$$m = 70 \text{ kg}$$

$$\Delta h = 3.0 \text{ m}$$

$$\Delta t = 2.0 \text{ s}$$

Solution

$$\Delta U = m \cdot g \cdot \Delta h$$

$$\Delta U = (70 \text{ kg})(9.81 \text{ m/s}^2)(3.0 \text{ m})$$

$$\Delta U \approx 2060 \text{ J}$$

And over 2.0 s, the average power on the centre of mass:

$$P = 2060 \text{ J} / 2.0 \text{ s}$$

$$P \approx 1030 \text{ W (about 1.4 horsepower)}$$

That 2060 J is the same number as the barbell lift earlier — a 100 kg bar raised 2.1 m and a 70 kg student raised 3.0 m are the same job. Once it is energy, it does not matter which.

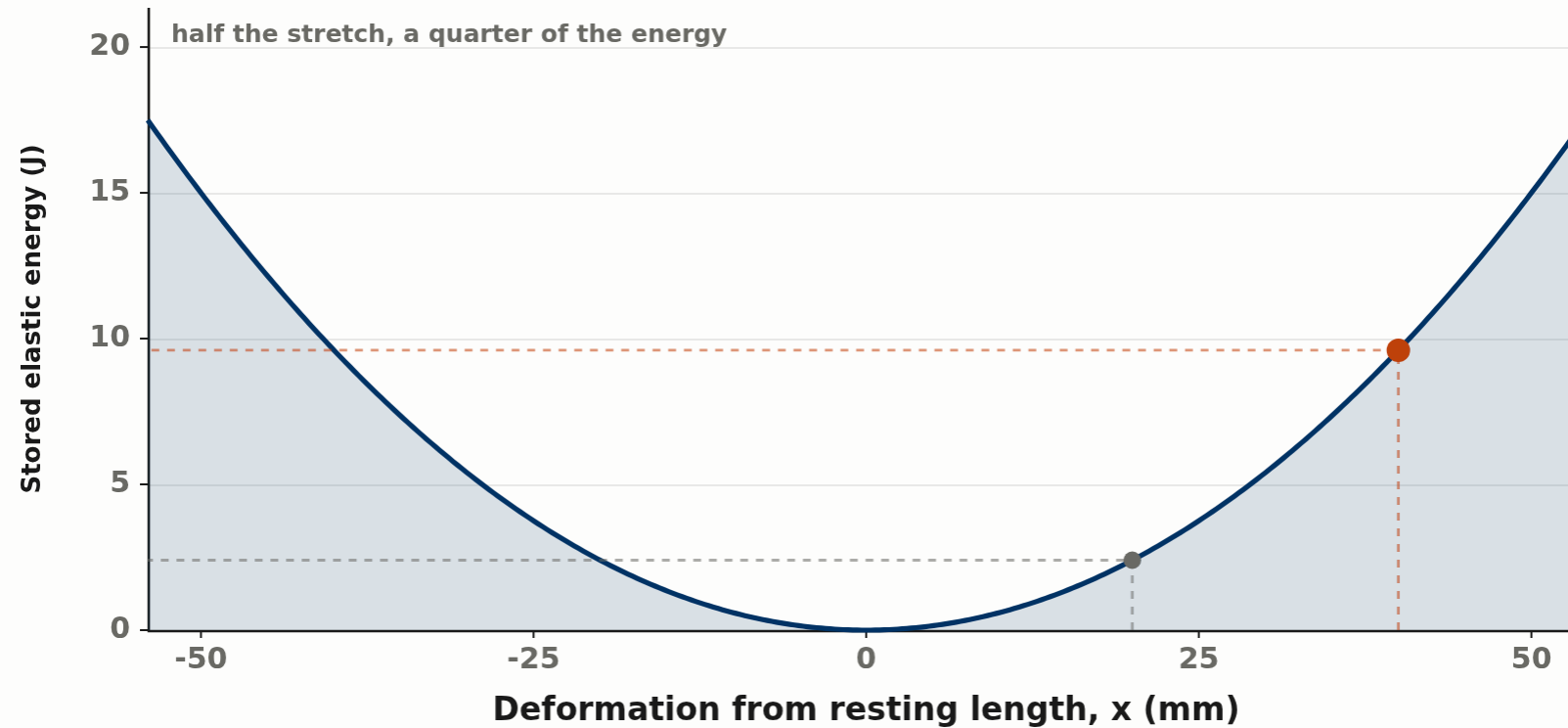
Energy stored in a deformed spring — or a tendon

$$U_e = \frac{1}{2} \cdot k \cdot x^2$$

- k is the stiffness, in N/m (or N/mm)
- **Double the stretch, four times the energy**
- x is the deformation from resting length
- Tendons, ligaments and aponeuroses are all biological springs

Elastic potential energy

Stiffness and stretch do not contribute equally



$$U = \frac{1}{2} \cdot k \cdot x^2 = \frac{1}{2} \times 12.0 \text{ N/mm} \times (40 \text{ mm})^2 = \mathbf{9.60 \text{ J}}$$

· at 20 mm it is only 2.40 J

elastic band

Achilles tendon

vaulting pole

Stiffness k



12.0
N/mm

Stretch x



40 mm

Conservation of mechanical energy

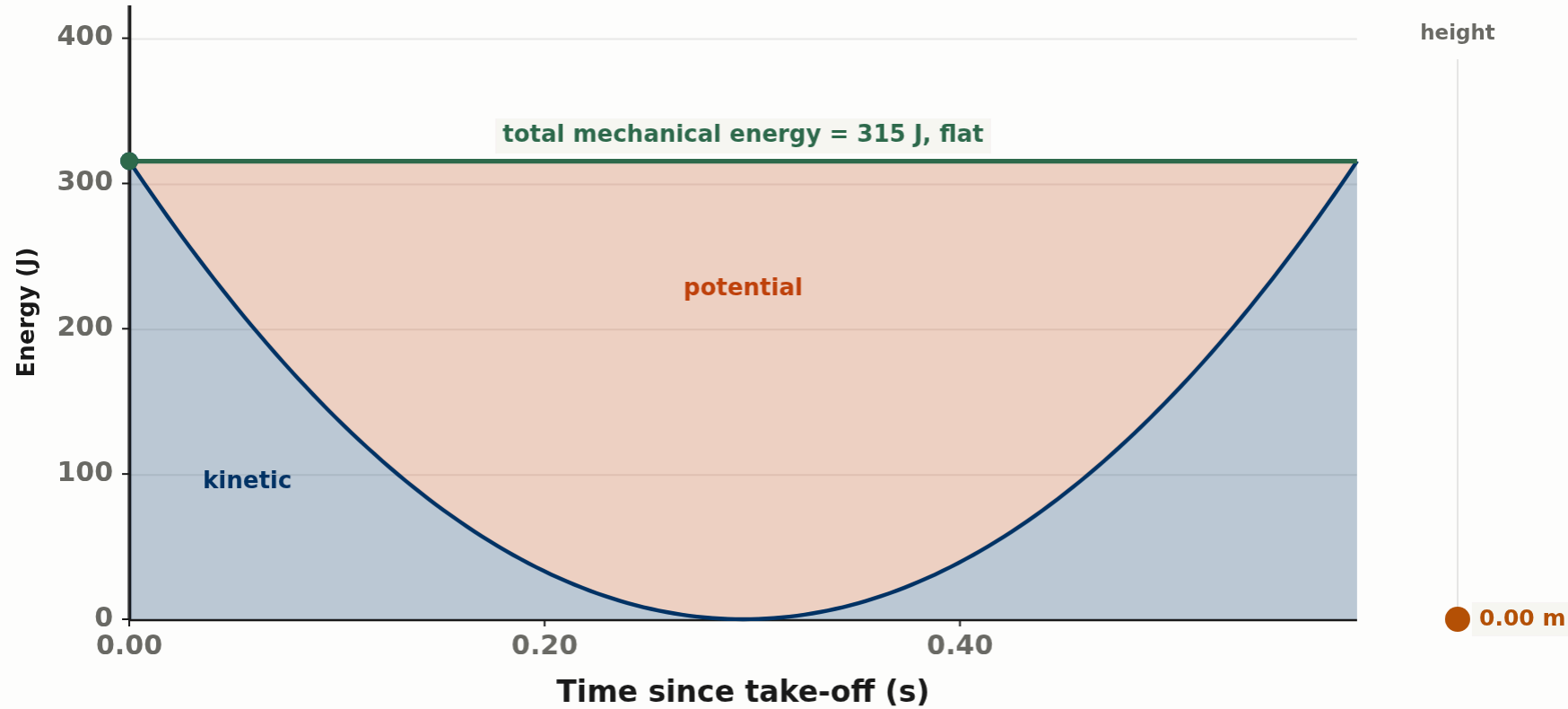
When only gravity and springs act, the total does not change

$$E_{\text{mech}} = KE + U_g + U_e = \text{constant}$$

- KE and PE trade back and forth, but their sum is flat
- This holds **only** in the absence of friction and muscle work
- Real movements lose some to friction and heat; muscles can add energy or take it away

Conservation of mechanical energy

Kinetic into potential and back again, through one flight



KE 315 J + PE 0 J = 315 J · peak rise $\Delta h = v^2/2g = 0.429$ m, whatever the mass

▶ Play

Through the
flight



0.00 s

Example Problem

A 75 kg athlete leaves the ground with a vertical velocity of 2.9 m/s. How high does their centre of mass rise?

Once they are airborne, only gravity acts — so mechanical energy is conserved.

Given:

$$m = 75 \text{ kg}$$

$$v_{\text{take-off}} = 2.9 \text{ m/s}$$

$$\text{at the peak, } v = 0$$

Solution

- At the peak all of the take-off kinetic energy has become potential energy

$$\frac{1}{2} \cdot m \cdot v_{\text{to}}^2 = m \cdot g \cdot \Delta h$$

$$\Delta h = v_{\text{to}}^2 / (2 \cdot g)$$

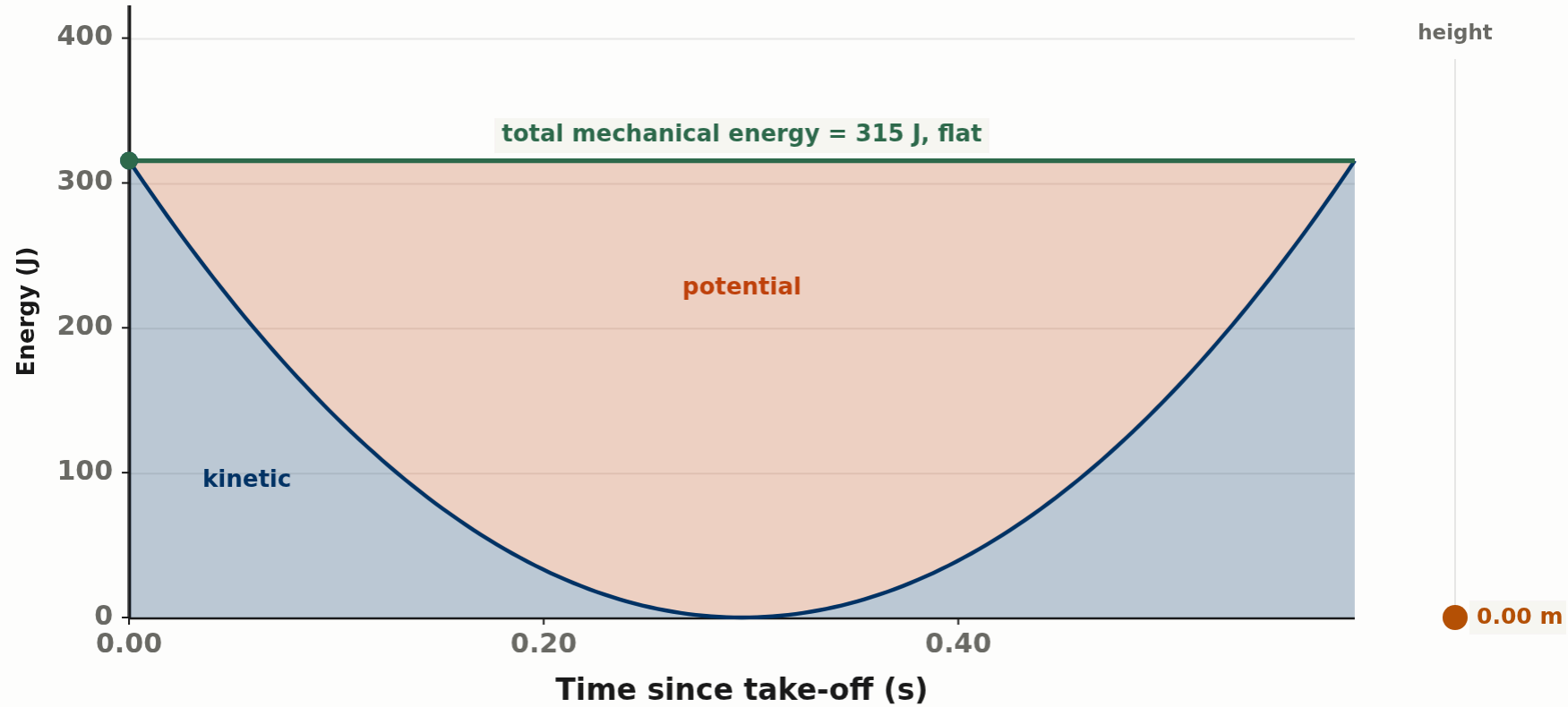
$$\Delta h = (2.9 \text{ m/s})^2 / (2 \times 9.81 \text{ m/s}^2)$$

$$\Delta h \approx 0.43 \text{ m (43 cm)}$$

The mass cancels. Jump height depends only on take-off velocity.

Conservation of mechanical energy

Try to make the mass matter



KE **315 J** + PE **0 J** = **315 J** · peak rise $\Delta h = v^2/2g = 0.429$ m, whatever the mass

Take-off speed m/s

Mass kg

Part 03

Power

The rate of mechanical work

Defining power

How fast work is done, or energy transferred

$$P = W / t$$

Time-average power — work done divided by the time it took

$$P = F \cdot v$$

Instantaneous linear power — force times velocity along the force

$$P = \tau \cdot \omega$$

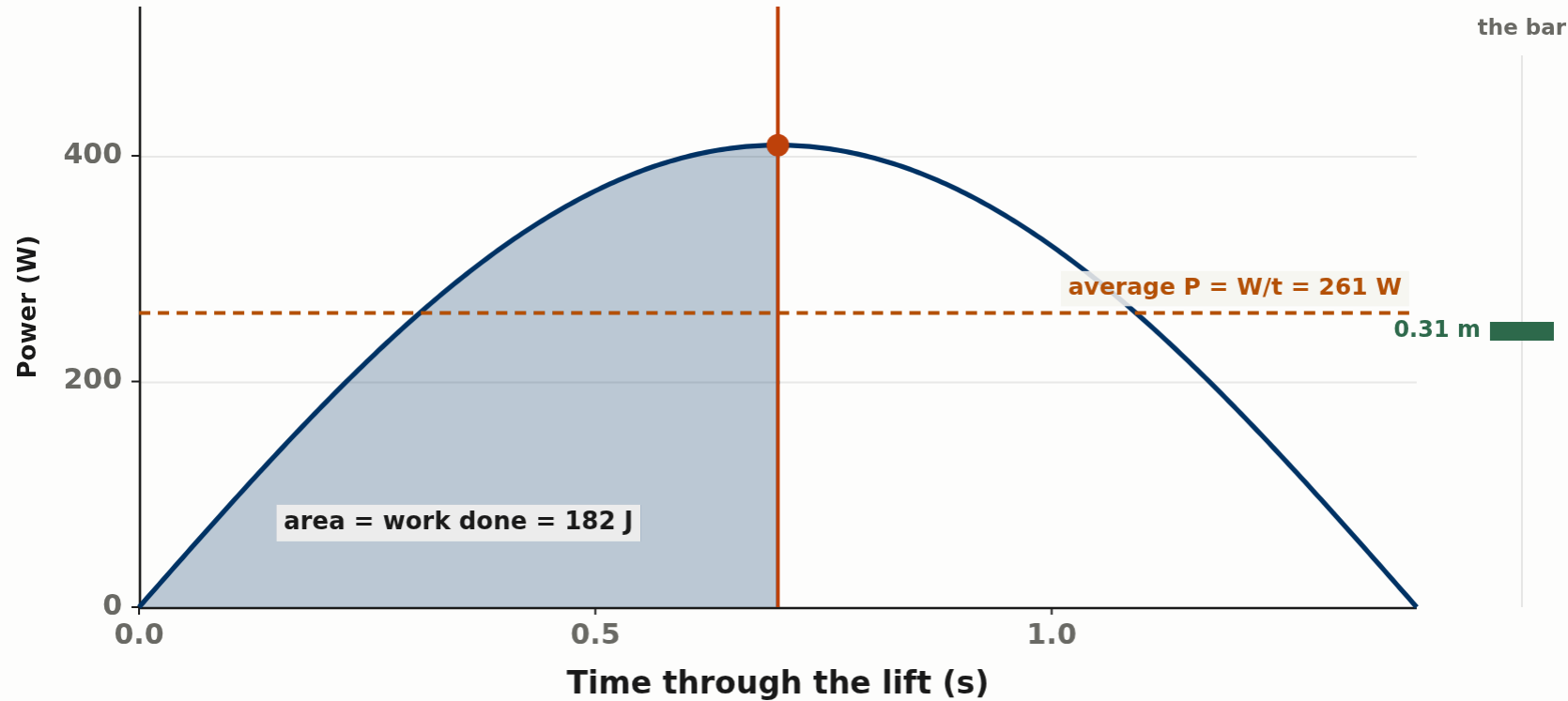
Rotational power — torque times angular velocity

SI unit: 1 watt (W) = 1 joule per second

For reference: 1 horsepower $\approx$ 746 W, and a fit cyclist can hold about 300 W for an hour.

Defining power

One lift, three formulas, the same watts



$$P = F \cdot v = 589 \text{ N} \times 0.70 \text{ m/s} = 409 \text{ W} \cdot W/t = 182 \text{ J} / 0.70 \text{ s} = 261 \text{ W}$$

Load

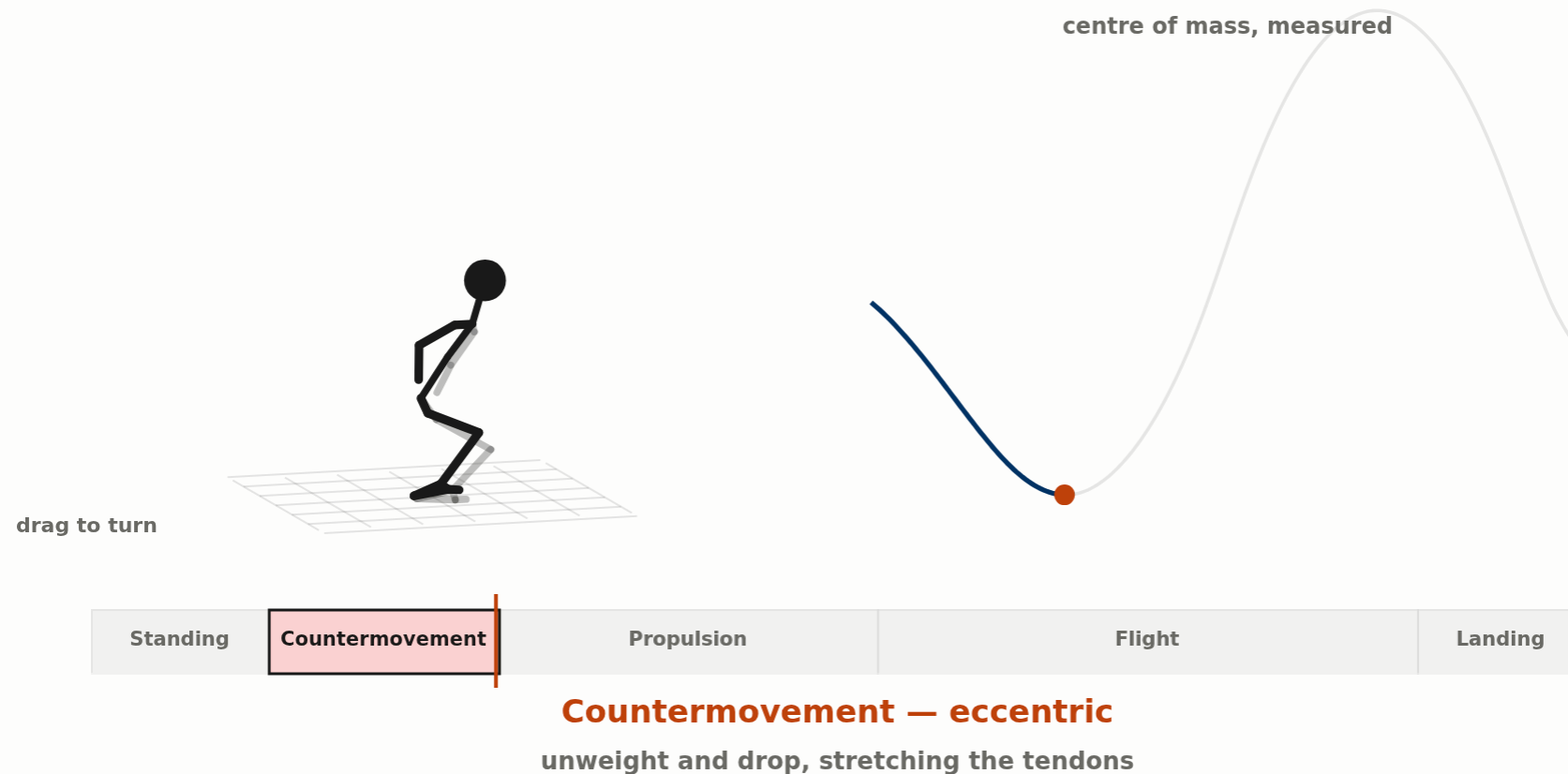
$P = F \cdot v$ $P = \tau \cdot \omega$

Through the lift

Lift takes

Phases of a countermovement jump

Each phase has its own work and power signature — drag the figure to turn it



Countermovement (eccentric) · unweight and drop, stretching the tendons · CoM -0.34 m from standing · SUSU-46, measured



Through the
jump



0.64 s

Example Problem — the Margaria-Kalamen stair sprint

A 75 kg athlete sprints up a staircase.

Timing gates are set on the 3rd and 9th steps, each step is 0.175 m high, and the athlete covers the six steps between the gates in 0.52 s.

What is their peak anaerobic power?

Given:

$$m = 75 \text{ kg}$$

$$\text{step} = 0.175 \text{ m, 6 steps}$$

$$\Delta t = 0.52 \text{ s}$$

Solution

- Only the **vertical** rise counts — the work being measured is against gravity

$$\Delta h = 6 \times 0.175 \text{ m} = 1.05 \text{ m}$$

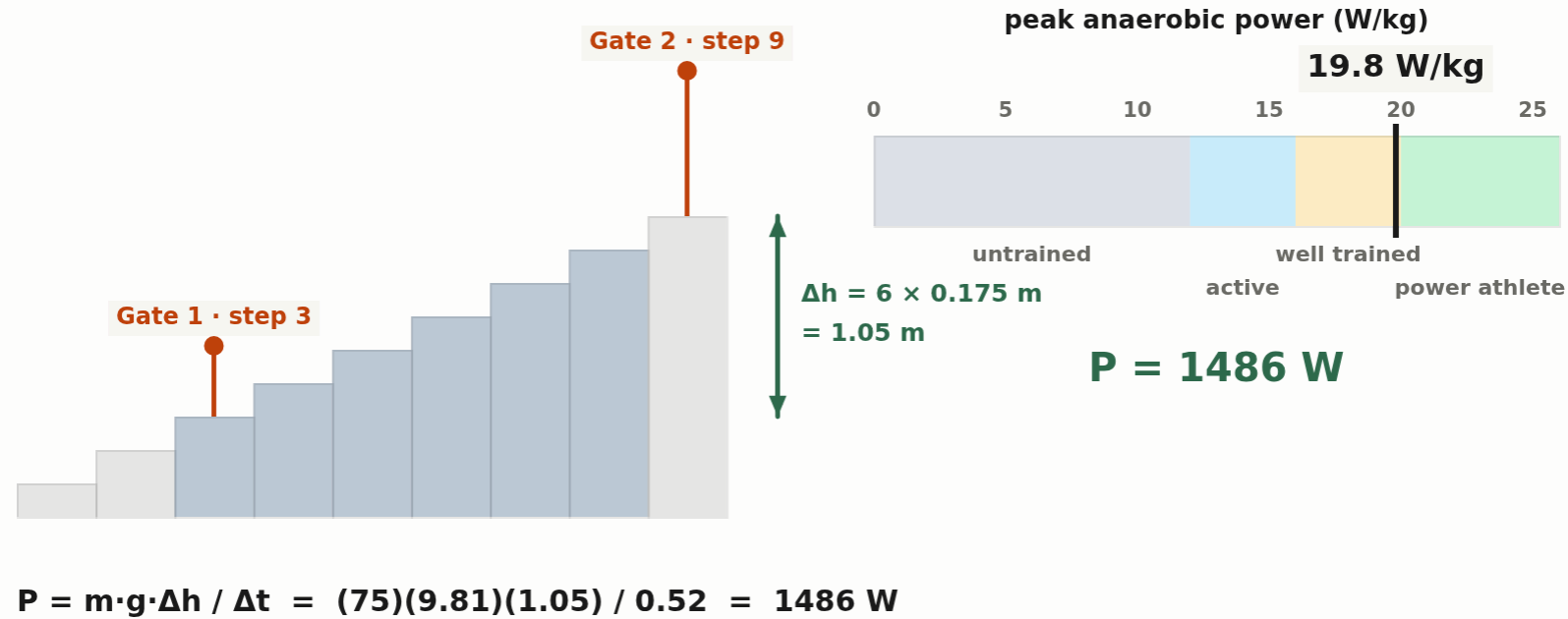
$$P = m \cdot g \cdot \Delta h / \Delta t$$

$$P = (75)(9.81)(1.05) / 0.52$$

$$P \approx 1486 \text{ W}$$

Per kilogram that is about **19.8 W/kg** — competitive for a power athlete.

The same test, with your own numbers in it



$$P = 1486 \text{ W} \cdot 19.8 \text{ W/kg} \cdot \Delta h = 1.05 \text{ m in } 0.52 \text{ s}$$

Mass

75 kg

Step
height

0.175
m

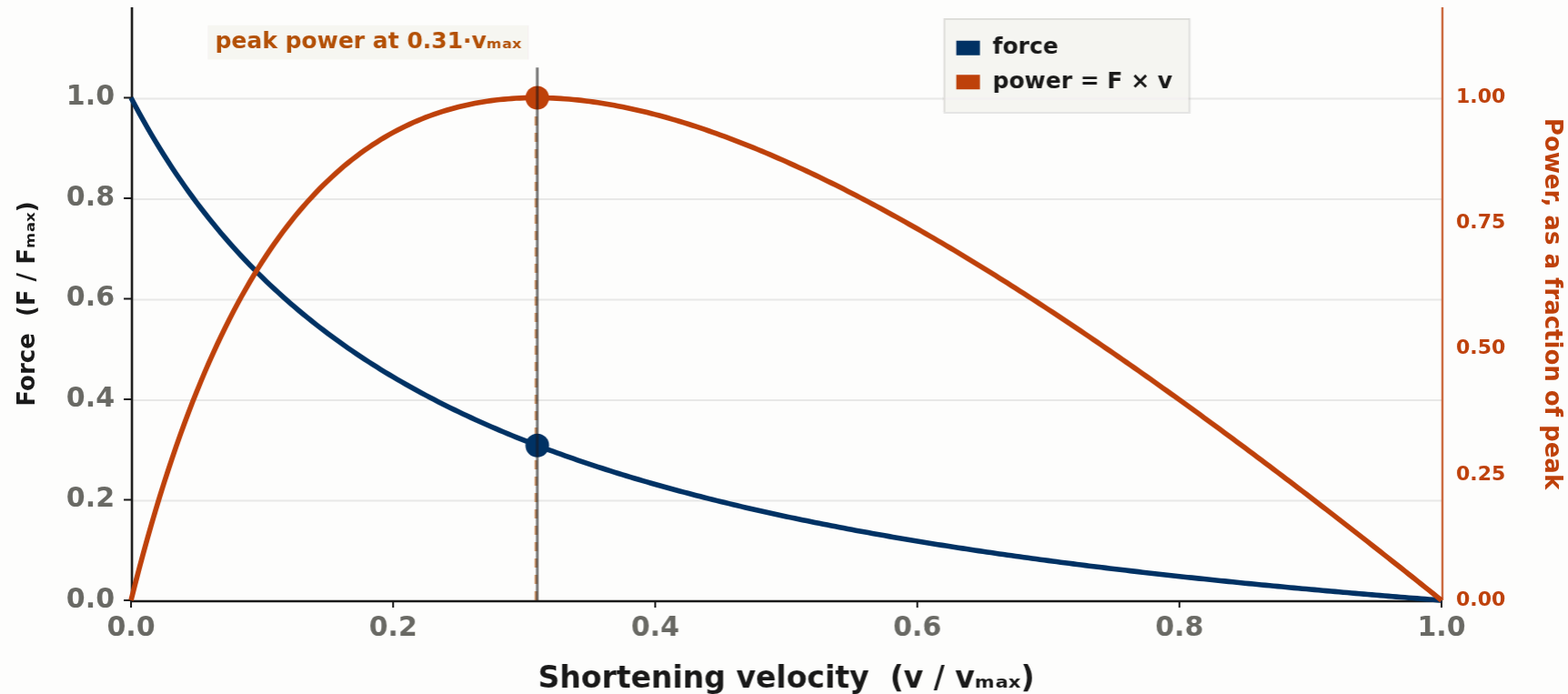
Peak power is at neither end of the range

- **Isometric** ($v = 0$): maximum force, and no power at all
- $v_{\max}$: no force, and again no power
- Peak power is at about $0.3 \cdot v_{\max}$
- Olympic lifts and plyometrics are aimed at that region

Power is a product, so it is zero whenever either term is zero. The maximum has to lie somewhere in between — and it is much closer to the fast end than most people guess.

Muscle force, velocity and power

Drag along the curve and watch the product



at $v = 0.31 \cdot v_{\max} \rightarrow F = 0.31 \cdot F_{\max} \cdot P = 1.00$ of peak · both terms matter — the product is what training targets

isometric ($v = 0$)

peak power

$v_{\max}$

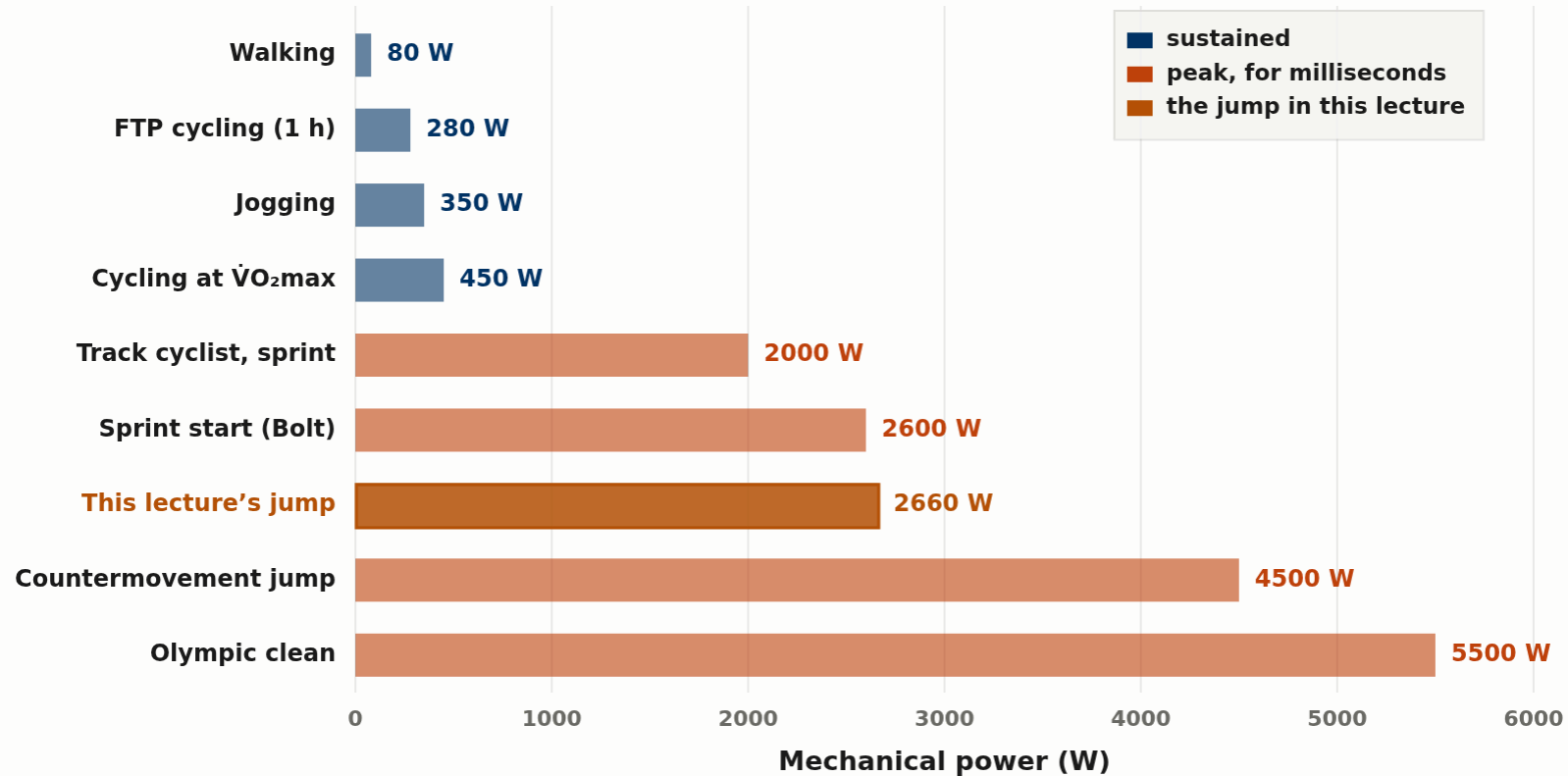
Shortening
velocity



$0.31 \cdot v_{\max}$

Peak power across activities

Explosive movements produce ten to fifty times steady-state power



Peak values last milliseconds — they set sprint starts, jump heights and throws. · a cyclist's watts are measured **at the pedal**; a jumper's are the **whole body**, so the two are not the same measurement

linear

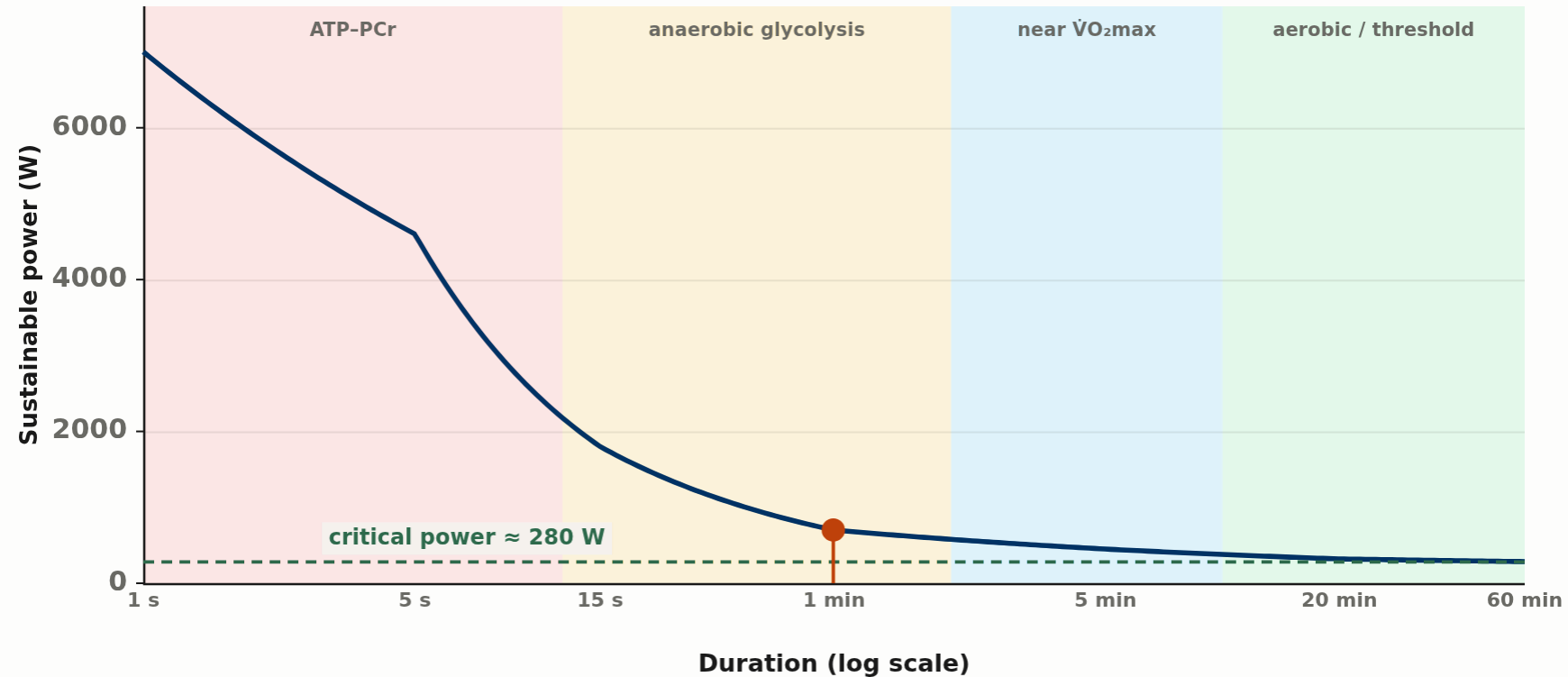
log

by size

sustained first

The power–duration curve

What you can hold falls sharply with how long you have to hold it



for **60 s** → about **702 W** · anaerobic glycolysis · 2.5× critical power

5 s sprint

1 min

5 min

60 min

Duration



60 s

Part 04

Applied biomechanics

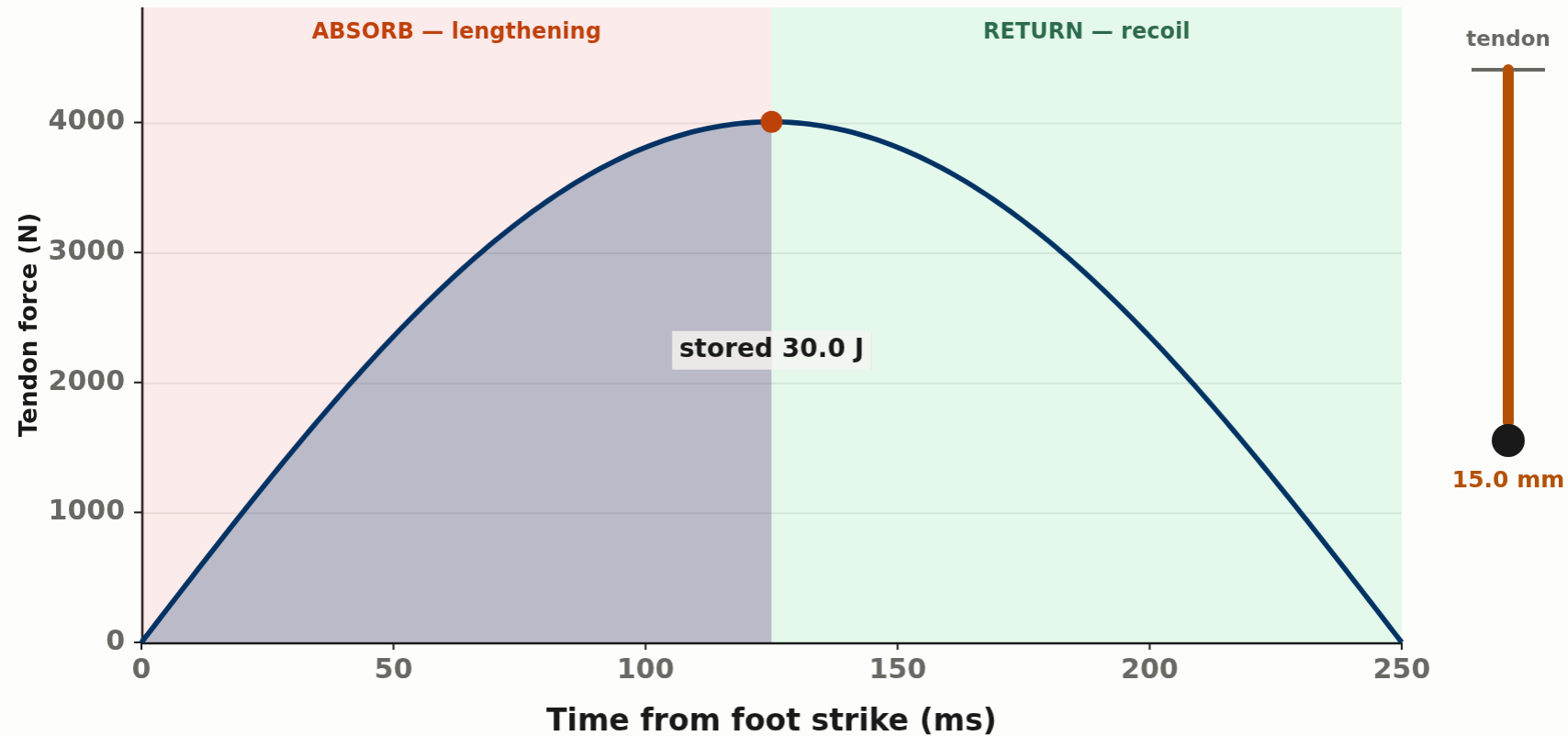
Stretch–shortening, jumping, gait

Tendons store and return energy on every step

- The Achilles returns roughly **35–50% of the mechanical energy a running step needs** — energy the muscles never have to produce
- Less muscle work means a lower metabolic cost
- It is why running is about 50% more economical than a muscle-only model predicts
- The same mechanism powers countermovement jumps and hopping

The stretch–shortening cycle

One running stance, absorbed and returned



peak stretch **15 mm** stores **30.0 J**; the tendon gives back **27.9 J** (93%) · about 40% of the ≈ 70 J a running step needs — energy the muscles never had to make



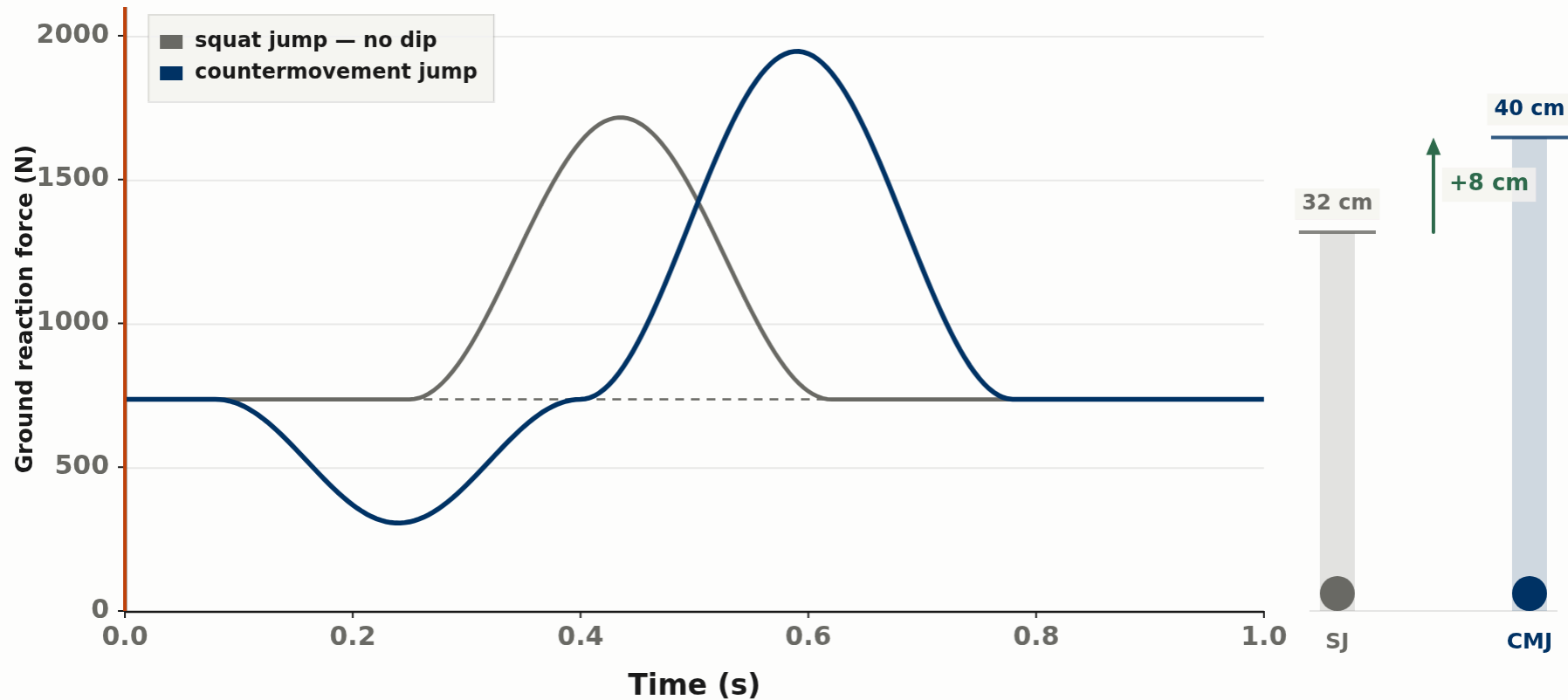
Through the
stance



125 ms

Countermovement against squat jump

Why the dip is worth 8 cm



squat jump **32 cm** · countermovement jump **40 cm** — **+8 cm** · elastic return, force already built before shortening, and a stretch reflex — all three, not one

▶ Play

Run both jumps



0.00 s

The same framework, different dominant terms

Sprinting	$W_{\text{net}} = \Delta KE$	Concentric work accelerates the runner; the stretch–shortening cycle recycles elastic energy each step
Vertical jump	$W_{\text{push-off}} = \Delta(KE + PE)$	Push-off work becomes take-off velocity, then height — with about 25% added by the countermovement
Cycling time trial	$P = F \cdot v$	Sustainable power at the pedal sets the pace, and the power–duration curve sets the ceiling
Throwing	$W = \Delta KE_{\text{implement}}$	Work is transferred down the kinetic chain and ends as kinetic energy in the hand and the implement
Running gait	$U_e \leftrightarrow KE$	Tendons and ground reaction impulses recycle energy across every step

WORK — $W = F \cdot d \cdot \cos \theta$

Positive, negative or zero depending on the angle between force and displacement. Measured in joules.

KINETIC ENERGY — $KE = \frac{1}{2}mv^2$

The v^2 term dominates. The work–energy theorem says $W_{\text{net}} = \Delta KE$.

POTENTIAL ENERGY — $U_g = mgh$, $U_e = \frac{1}{2}kx^2$

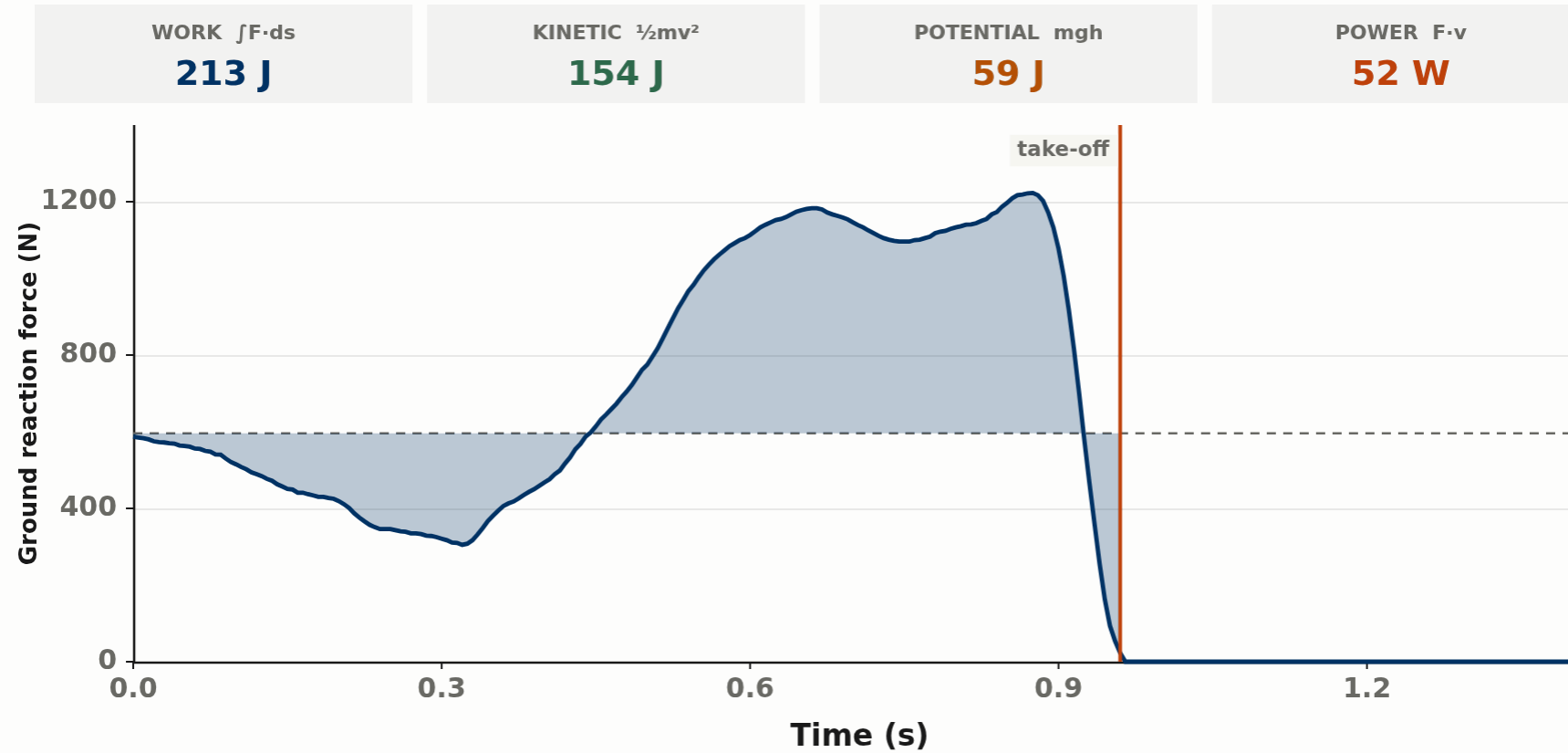
Gravitational depends on a reference you choose; elastic is quadratic in the stretch. Both trade with kinetic energy.

POWER — $P = W/t = F \cdot v = \tau \cdot \omega$

The rate of doing work. Muscle power peaks near $0.3 \cdot v_{\text{max}}$, and what you can sustain falls sharply with duration.

One movement, four quantities

The whole lecture read off a single jump



one measured jump, four quantities · $v = 2.25 \text{ m/s}$ · $W = \Delta KE + \Delta PE$ at every instant, which is the whole lecture in one line

Through the
jump



0.96 s

EPHE 341 · BIOMECHANICS

Questions?

Next up: angular kinematics · Further
reading: McGinnis Chpt 4